

Financial News Sentiment vs. NIFTY50 Movement

Project Report

1 Project Overview

This project investigates whether sentiment expressed in financial news headlines can help explain or predict the direction of the NIFTY50 index. The analysis combines financial news data with historical NIFTY50 market data and develops two sentiment-analysis approaches: a rule-based financial sentiment method and the pretrained FinBERT financial-language model.

The complete workflow includes data cleaning, date alignment, news aggregation, rule-based sentiment scoring, FinBERT sentiment extraction, feature engineering, exploratory analysis, time-aware machine learning, model comparison, prediction-error analysis, and a simple backtest.

The central objective is to determine whether financial-news sentiment, when combined with recent market information, provides a useful signal for predicting the next-day direction of the NIFTY50.

Key Results	
Merged observations	132
Usable ML observations	125
Training / Test split	100 / 25
Rule-based same-day accuracy	59.0%
Best original test accuracy	44.0%
Best FinBERT test accuracy	36.0%
Best CV accuracy	51.25%
Rule-Based vs. FinBERT correlation	0.407
Buy-and-Hold return	-0.64%
Model strategy return	-0.77%

2 Changes and Contributions

The project was extended beyond the original rule-based sentiment pipeline by introducing FinBERT and a second machine-learning feature set.

The main changes and contributions were:

- Cleaned and standardized the financial news and NIFTY50 datasets.
- Aligned financial news and market observations using trading dates.
- Grouped multiple news headlines occurring on the same date.
- Implemented a financial rule-based sentiment scoring system using positive and negative financial vocabulary.
- Constructed a same-day sentiment-based prediction rule and evaluated its accuracy.
- Corrected the machine-learning target so that models predict the **next trading day's** NIFTY50 direction.
- Created rolling sentiment features including three-day and seven-day moving averages and short-term sentiment variability.
- Created market features including lagged movement, volume change, and high-low spread.
- Added FinBERT using the `ProsusAI/finbert` pretrained financial-language model.
- Converted FinBERT predictions into numerical sentiment scores using confidence-weighted positive and negative values.
- Created FinBERT rolling features: `finbert_ma3`, `finbert_ma7`, and `finbert_std3`.
- Compared rule-based sentiment directly with FinBERT sentiment.
- Measured the correlation between the two sentiment approaches.
- Trained Logistic Regression and Random Forest models using the FinBERT-based feature set.
- Compared the original models and FinBERT-based models on the same chronological test period.
- Used a chronological train/test split to avoid future-data leakage.

- Performed Random Forest hyperparameter tuning using TimeSeriesSplit cross-validation.
- Evaluated predictions using accuracy, classification reports, and confusion matrices.
- Analyzed Random Forest feature importance.
- Compared a model-based trading strategy against buy-and-hold.

3 Data and Feature Engineering

The financial news dataset contains multiple headlines per trading day, while the NIFTY50 dataset contains daily OHLC and trading information. The two datasets were cleaned and aligned using their trading dates.

After merging the datasets, 132 observations were available. Following the creation of rolling features and removal of rows containing unavailable values, 125 observations were used for the final machine-learning experiments.

The original feature set consisted of eight features:

sentiment_score	sentiment_ma3	sentiment_ma7	sentiment_std3
movement_lag1	movement_lag2	volume_change	high_low_spread

The FinBERT feature set replaced the original sentiment features with four FinBERT-derived variables while retaining the market features:

finbert_score	finbert_ma3	finbert_ma7	finbert_std3
movement_lag1	movement_lag2	volume_change	high_low_spread

The target variable represents whether the NIFTY50 moves upward on the following trading day. The data was sorted chronologically before the target was shifted.

The final machine-learning dataset contained 125 observations, with 100 observations used for training and 25 observations reserved for the final test period.

Time-aware evaluation: The first 100 observations were used for training, while the final 25 observations were kept completely unseen for final evaluation. This chronological design prevents future observations from being mixed into the training data.

4 Rule-Based Sentiment Analysis

The original sentiment system assigns scores to financial news headlines using a manually constructed financial vocabulary. Positive terms increase the score while negative terms decrease it.

The sentiment score is then converted into a simple prediction:

- Positive score → UP
- Negative score → DOWN
- Zero score → NEUTRAL

For the binary prediction used in evaluation, positive sentiment was mapped to an upward movement and non-positive sentiment to a downward movement.

The rule-based approach achieved a same-day prediction accuracy of approximately 59.0%.

However, this result must not be interpreted as a next-day forecasting result. The rule-based sentiment score and the original target are calculated from the same trading day. Therefore, the 59.0% accuracy primarily reflects same-day co-movement between news sentiment and market movement rather than genuine future prediction.

This distinction motivated the next-day target used in the machine learning section.

5 Original Exploratory Data Analysis

The original rule-based sentiment distribution is shown in Figure 1.

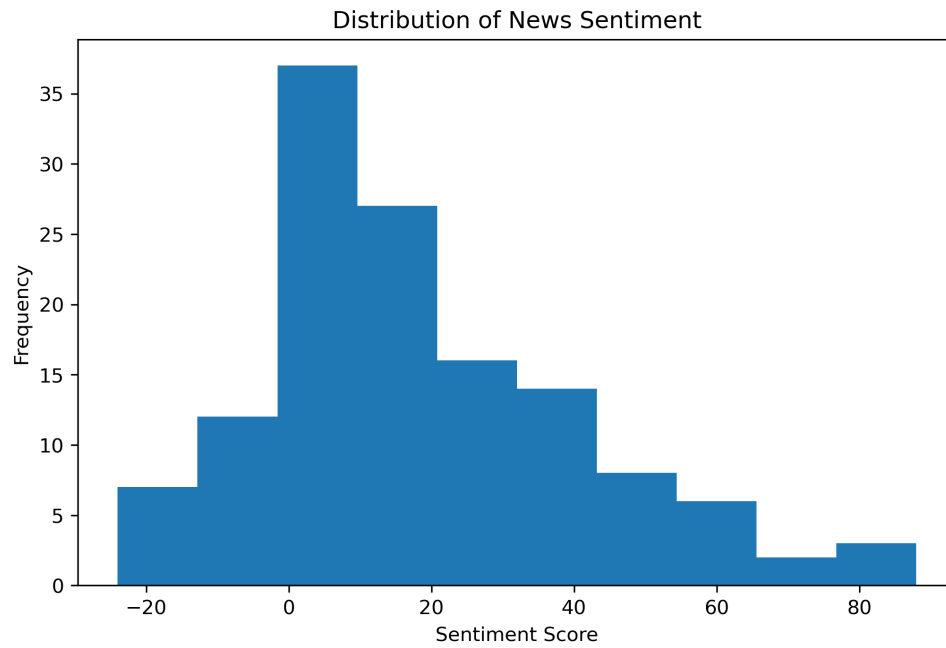


Figure 1. Distribution of the original rule-based financial-news sentiment scores.

The average rule-based sentiment by market direction is shown in Figure 2.

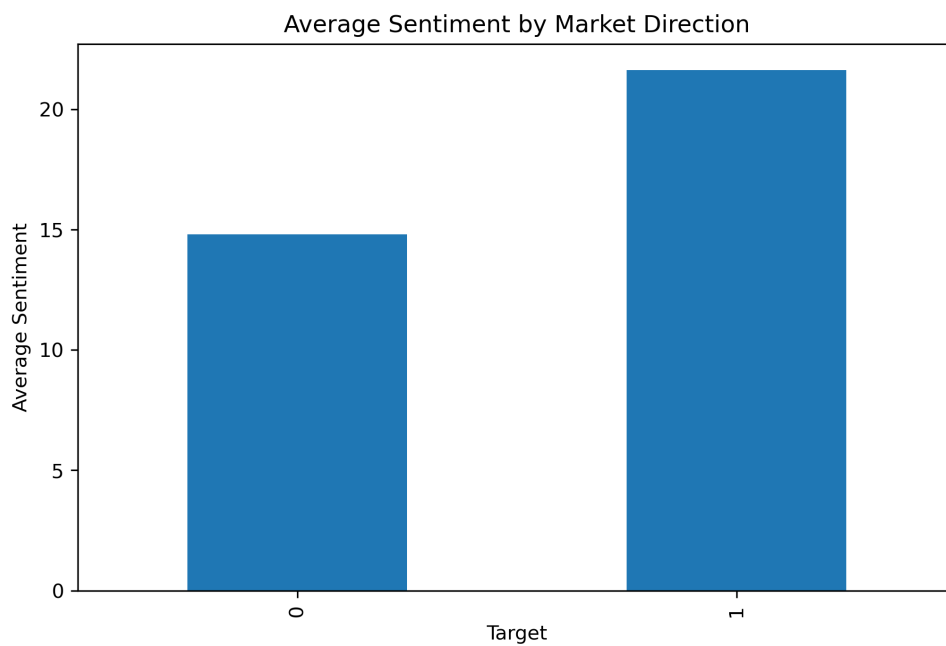


Figure 2. Average rule-based sentiment by NIFTY50 market direction.

The relationship between rule-based sentiment and same-day market movement is shown in Figure 3.

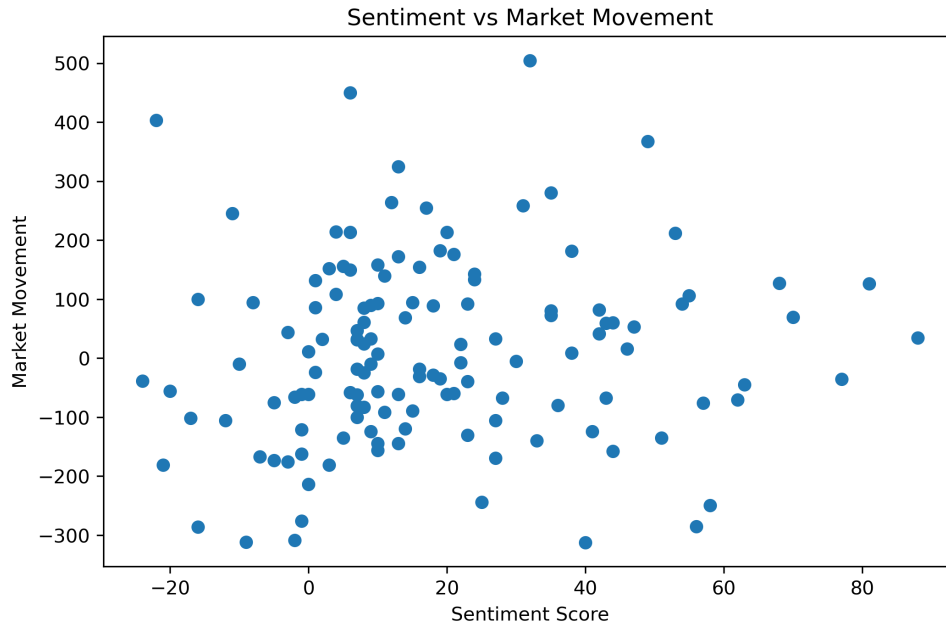


Figure 3. Relationship between rule-based sentiment and same-day NIFTY50 market movement.

These plots provide useful exploratory evidence that financial-news sentiment and market movement can co-move. They should not, however, be interpreted as evidence of reliable next-day prediction.

6 FinBERT Sentiment Analysis

To extend the original rule-based approach, the project introduced FinBERT, a pretrained language model designed for financial text.

Each headline was processed using FinBERT to obtain:

- a sentiment label: positive, neutral, or negative;
- a confidence score;
- a numerical sentiment score.

The numerical FinBERT sentiment score was constructed as follows:

$$S_{\text{FinBERT}} = \begin{cases} +C, & \text{if the label is positive,} \\ -C, & \text{if the label is negative,} \\ 0, & \text{if the label is neutral,} \end{cases}$$

where C represents the model confidence.

This produces a continuous sentiment score that preserves both the direction and confidence of the FinBERT prediction.

6.1 FinBERT Sentiment Distribution

The distribution of FinBERT sentiment labels is shown below.

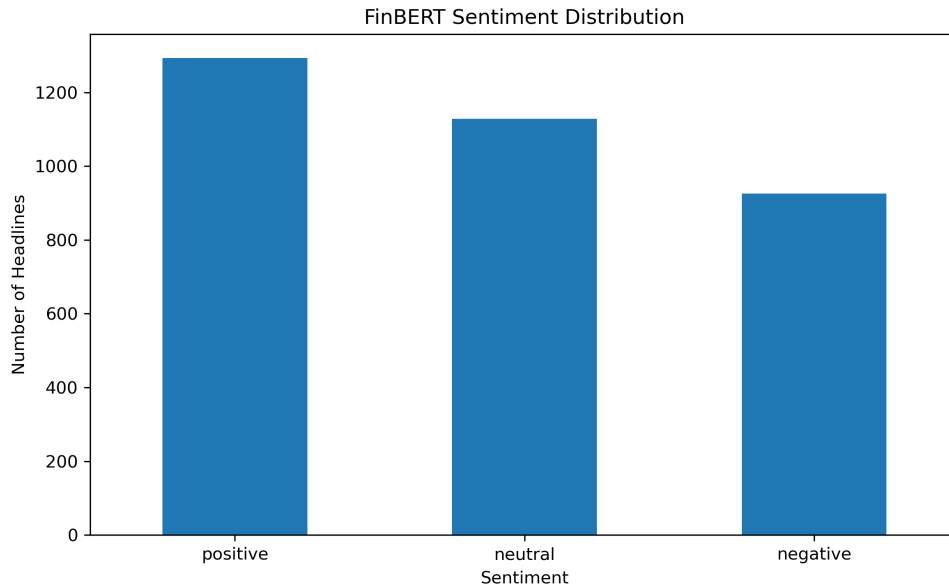


Figure 4. Distribution of FinBERT sentiment labels across financial-news headlines.

The FinBERT output provides a model-based alternative to the manually constructed rule-based vocabulary. Unlike the rule-based approach, FinBERT evaluates the linguistic context of each headline using a pretrained financial-language model.

6.2 Rule-Based Sentiment vs. FinBERT

The two sentiment approaches were directly compared using their numerical scores.

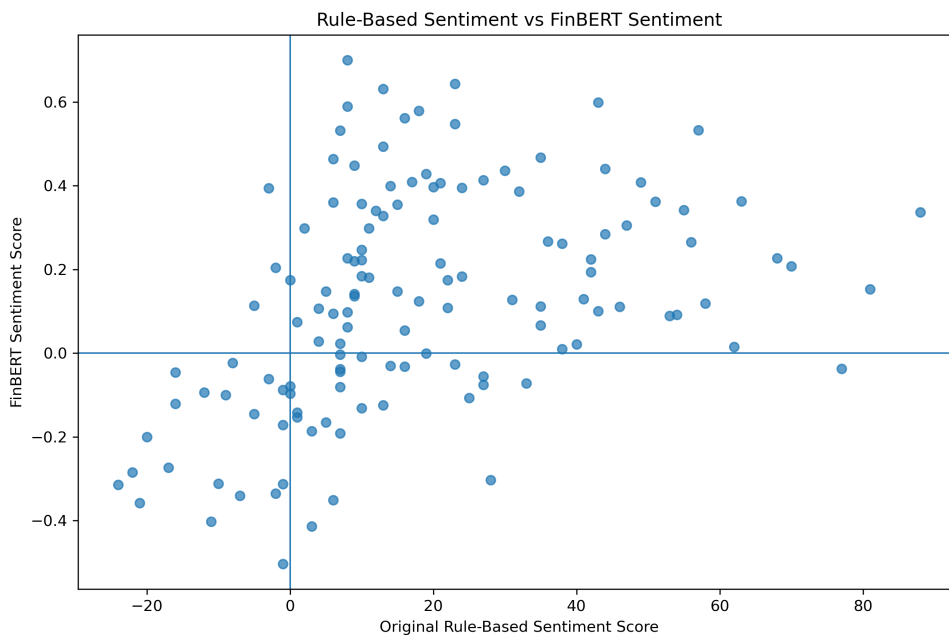


Figure 5. Comparison between the original rule-based sentiment score and the FinBERT sentiment score.

The correlation between the two sentiment measures was approximately 0.407.

This represents a moderate positive relationship: the two methods often identify broadly similar sentiment direction, but they are far from identical. This is expected because the rule-based method relies on manually specified financial vocabulary, whereas FinBERT uses a pretrained language representation.

The comparison therefore demonstrates that FinBERT provides additional information rather than simply reproducing the original rule-based score.

Important interpretation: A correlation of 0.407 indicates meaningful overlap between the two sentiment measures, but it is not sufficiently high to consider them equivalent. The methods capture different aspects of headline sentiment.

7 Machine Learning Analysis

The machine-learning task was framed as a next-day classification problem. The target was shifted by one trading day so that features available on day t were used to predict the market direction on day $t + 1$.

The original models evaluated were:

- Logistic Regression
- Random Forest
- XGBoost
- Tuned Random Forest

The FinBERT feature set was subsequently evaluated using:

- FinBERT Logistic Regression
- FinBERT Random Forest

A majority-class classifier was included as the baseline.

7.1 Original Model Results

Table 1. Original model performance on the held-out test period.

Model	Accuracy
Majority Baseline	44.0%
Logistic Regression	44.0%
Random Forest	44.0%
XGBoost	40.0%
Tuned Random Forest	44.0%

The original Logistic Regression and Random Forest models matched the 44.0% majority baseline. XGBoost performed slightly worse at 40.0%.

The tuned Random Forest selected the following parameters:

n_estimators	100
max_depth	3
min_samples_leaf	20

The best time-series cross-validation accuracy was 51.25%, but the accuracy on the untouched test period remained 44.0%.

Therefore, hyperparameter tuning improved the cross-validation selection process but did not improve final out-of-sample accuracy.

7.2 FinBERT Model Results

The FinBERT-based machine-learning dataset used the four FinBERT sentiment features together with four market features.

The FinBERT model results were:

Table 2. FinBERT-based model performance on the held-out test period.

Model	Accuracy
FinBERT Logistic Regression	36.0%
FinBERT Random Forest	28.0%

The FinBERT Logistic Regression model achieved 36.0% accuracy, while the FinBERT Random Forest achieved 28.0%. Therefore, replacing the original rule-based sentiment features with the FinBERT-derived features did not improve predictive performance on the final test period.

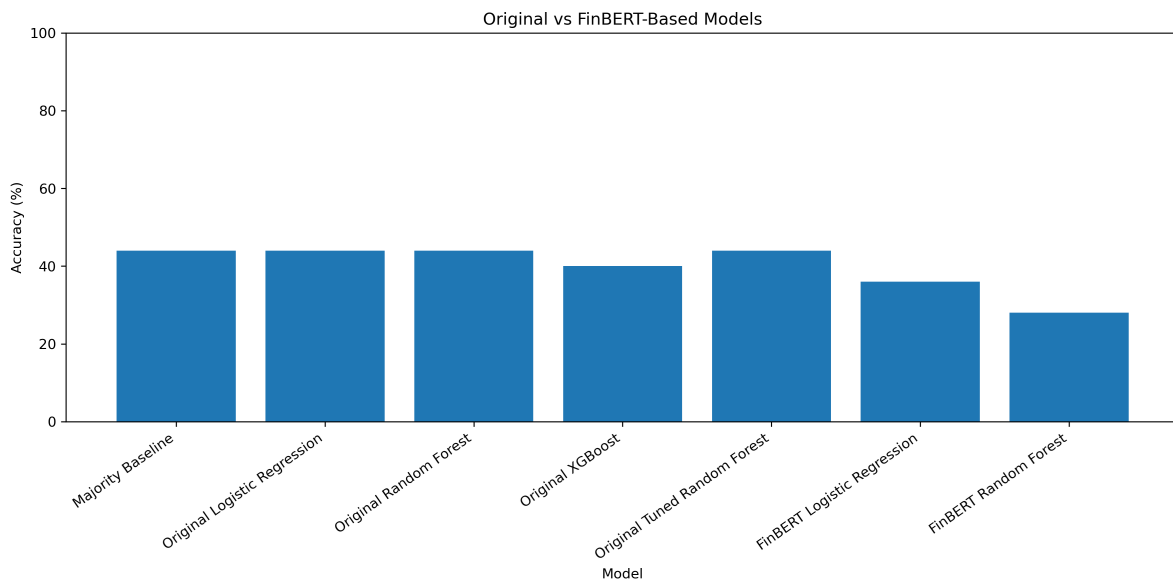
8 Final Model Comparison

The complete comparison between the original and FinBERT-based approaches is shown in Table 3.

Table 3. Complete model comparison on the held-out test period.

Model	Accuracy
Majority Baseline	44.0%
Original Logistic Regression	44.0%
Original Random Forest	44.0%
Original XGBoost	40.0%
Original Tuned Random Forest	44.0%
FinBERT Logistic Regression	36.0%
FinBERT Random Forest	28.0%

The complete comparison is also visualized in Figure 6.

**Figure 6.** Final comparison of original and FinBERT-based machine-learning models.

The most important result is that the original tuned Random Forest remained the strongest model on the held-out test set at 44.0%. FinBERT Logistic Regression reached 36.0%, while FinBERT Random Forest reached only 28.0%. This means that the addition of FinBERT did not automatically translate into better next-day prediction. In this dataset, recent market features and the original sentiment representation were at least as useful as the FinBERT representation for the tested models.

9 Feature Importance

The tuned Random Forest feature importance plot provides an indication of which variables contributed most strongly to the model's decisions.

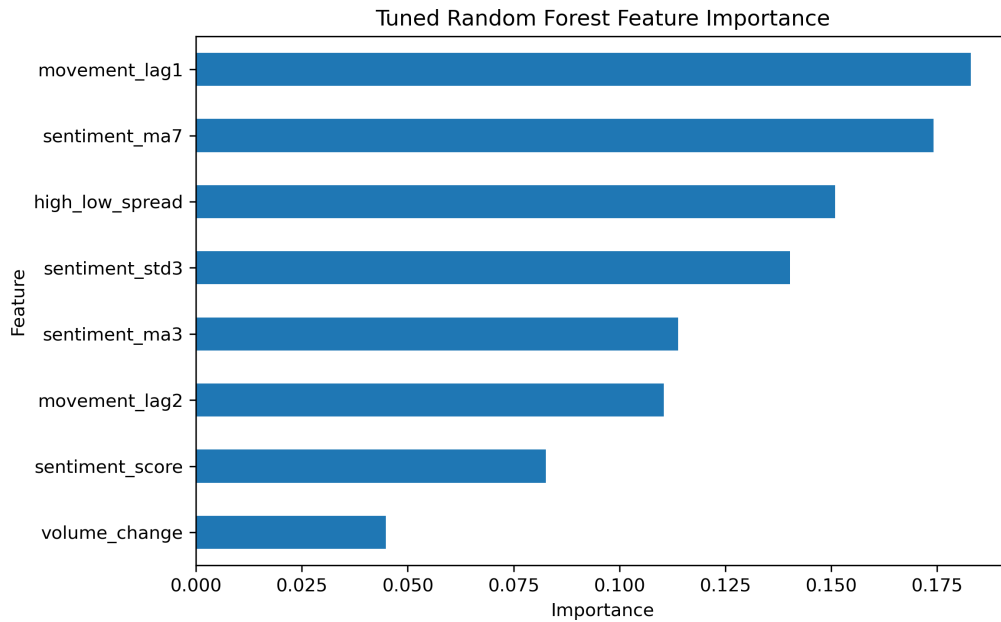


Figure 7. Feature importance from the tuned Random Forest.

Previous-day market movement was among the most influential features, while sentiment-derived variables also contributed to the model.

These results should be interpreted as model feature importance rather than causal evidence. A high feature-importance value does not mean that the feature independently causes future market movement.

The results suggest that recent price behaviour contains substantial predictive information relative to the sentiment features in this small dataset.

10 Prediction Errors

The tuned Random Forest produced the following confusion matrix on the 25-observation test period.

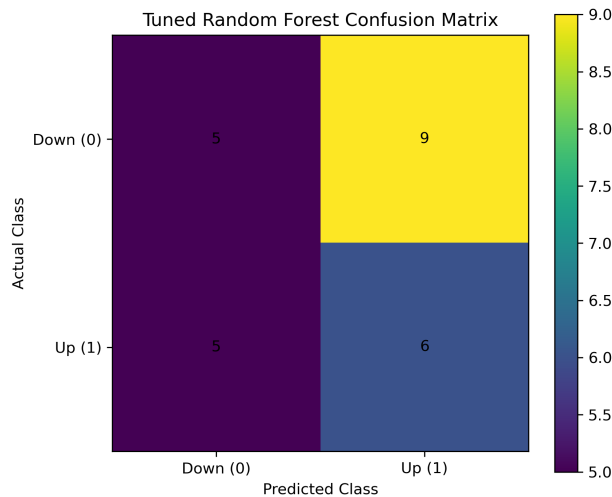


Figure 8. Confusion matrix for the tuned Random Forest.

The model correctly classified 5 downward movements and 6 upward movements, resulting in 11 correct predictions out of 25 test observations.

Because the test set contains only 25 observations, one additional correct prediction changes the measured accuracy by approximately four percentage points. This illustrates why the reported test accuracy should not be treated as a highly precise estimate of long-term predictive performance.

11 Backtest and Discussion

A simple long-only strategy was constructed using the tuned Random Forest predictions. When the model predicted an upward movement, the strategy held the NIFTY50 for the following day; otherwise, it remained out of the market.

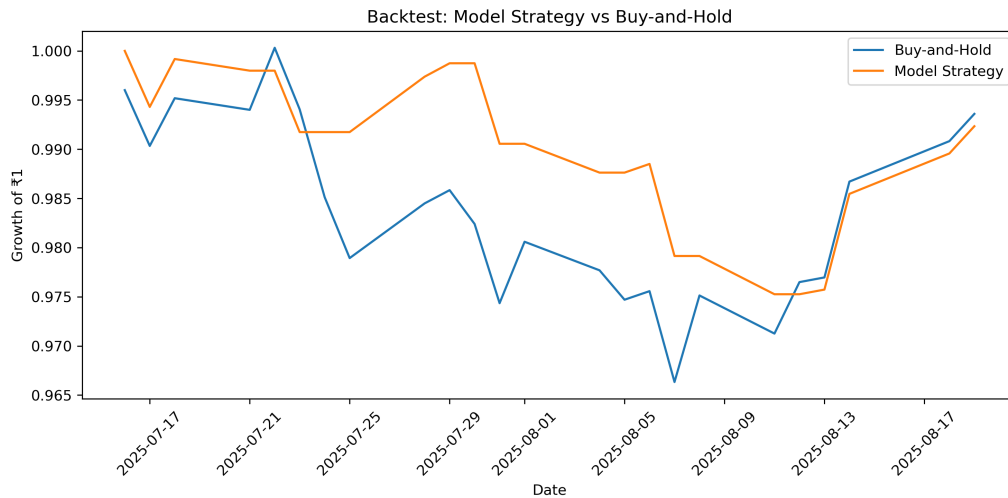


Figure 9. Model strategy compared with buy-and-hold during the test period.

Over the evaluation period, the buy-and-hold strategy returned -0.64% , while the model-based strategy returned -0.77% .

Therefore, the model strategy underperformed buy-and-hold by approximately 0.13 percentage points.

The backtest is intended only as an illustrative evaluation. It does not include transaction costs, slippage, taxes, or other practical trading considerations. The small test window also means that the result should not be interpreted as evidence for or against a deployable trading strategy.

12 Discussion of the FinBERT Extension

The FinBERT extension provides an important comparison with the original rule-based methodology.

The rule-based system is simple, transparent, and easy to interpret. Its score is directly determined by the occurrence of manually selected financial terms. FinBERT, in contrast, uses a pretrained language model that evaluates financial language using contextual representations.

The moderate correlation of 0.407 between the two approaches indicates that they capture some common sentiment information while still producing substantially different scores.

However, the machine-learning results show that a more sophisticated sentiment representation does not necessarily produce better next-day market prediction.

The original tuned Random Forest achieved 44.0%, compared with 36.0% for FinBERT Logistic Regression and 28.0% for FinBERT Random Forest. All of these results were obtained using the same chronological held-out test period.

This is an important negative result rather than a failure of the project. It demonstrates that improving the sophistication of the NLP component does not automatically improve financial forecasting performance, particularly when the dataset is small and market movement is noisy.

13 Limitations

Several limitations should be considered when interpreting the results.

- The machine-learning dataset contains only 125 usable observations.
- The final test period contains only 25 observations.
- The small sample size makes the reported accuracy sensitive to individual predictions.
- The rule-based 59.0% result is a same-day co-movement measurement and should not be interpreted as a future forecasting accuracy.
- The FinBERT model was used as a pretrained model rather than fine-tuned specifically on the NIFTY50 news dataset.
- The FinBERT sentiment features were derived from headline sentiment rather than full article text.
- The dataset covers a relatively short financial period.

- The backtest does not include transaction costs, slippage, taxes, or market-impact effects.
- Feature importance measures model dependence and should not be interpreted as causal relationships.
- The moderate correlation between rule-based and FinBERT sentiment does not imply that either method is a reliable predictor of future returns.

14 Conclusion

The analysis investigated whether financial-news sentiment could help predict next-day NIFTY50 direction.

The original rule-based sentiment system achieved 59.0% accuracy when evaluated against the same-day target. However, this result reflects same-day co-movement rather than genuine future prediction.

For the correctly framed next-day prediction problem, the strongest original model was the tuned Random Forest, which achieved 44.0% accuracy on the held-out test period. This was equal to the majority-class baseline.

The FinBERT extension produced a 36.0% accuracy for Logistic Regression and 28.0% for Random Forest. Consequently, FinBERT did not improve the predictive performance of the tested models.

The comparison between rule-based sentiment and FinBERT produced a correlation of 0.407, demonstrating that the two approaches capture related but distinct aspects of financial-news sentiment.

The backtest also showed that the model-based strategy returned -0.77% , compared with -0.64% for buy-and-hold during the test period.

Overall conclusion:

The tested sentiment representations and machine-learning models did not outperform the majority baseline on the held-out next-day prediction task. FinBERT provided a more sophisticated contextual sentiment representation, but it did not translate into improved predictive accuracy in this small dataset.

Nevertheless, the project demonstrates a complete time-aware financial machine-learning workflow: data cleaning, sentiment extraction, feature engineering, next-day target construction, chronological evaluation, NLP model comparison, feature analysis, and strategy backtesting.

15 Final Summary

The final pipeline can be summarized as:

Financial News → Cleaning → Date Alignment → Rule-Based Sentiment → FinBERT Sentiment → Feature Engineering → Next-Day Target → Time-Aware ML → Model Comparison → Backtest

The final comparison therefore answers the original research question conservatively: financial-news sentiment contains information related to market movement, but the tested sentiment and machine-learning approaches did not provide reliable evidence of improved next-day NIFTY50 prediction on the held-out period.